

Practical 03

Question

Develop a C program using `fork()` that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution.

```
GNU nano 7.2 week3_task3.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

void display_process(const char *stage)
{
    printf("[%s] PID = %d, PPID = %d\n",
           stage, getpid(), getppid());
}

int main()
{
    pid_t pid;

    printf("Parent process started.\n");
    display_process("Parent - Before fork");

    pid = fork();
    if (pid < 0)
    {
        perror("fork failed");
        return 1;
    }
    if (pid == 0)
    {
        // Child process
        display_process("Child - Created");
        printf("Child process is running...\n");
        sleep(3);
        display_process("Child - After sleep");
        printf("Child process is terminating.\n");
        exit(0);
    }
    else
    {
        // Parent process
        display_process("Parent - After fork");
        printf("Parent is waiting for child...\n");
        sleep(1);
        display_process("Parent - Waiting");
        wait(NULL);
        display_process("Parent - After child termination");
        printf("Parent process is terminating.\n");
    }
    return 0;
}
```

```
akhil@DESKTOP-HL1VUUS:~$ nano week3_task3.c
akhil@DESKTOP-HL1VUUS:~$ gcc week3_task3.c -o week3_task3
akhil@DESKTOP-HL1VUUS:~$ ./week3_task3
Parent process started.
[Parent - Before fork] PID = 524, PPID = 456
[Parent - After fork] PID = 524, PPID = 456
Parent is waiting for child...
[Child - Created] PID = 525, PPID = 524
Child process is running...
[Parent - Waiting] PID = 524, PPID = 456
[Child - After sleep] PID = 525, PPID = 524
Child process is terminating.
[Parent - After child termination] PID = 524, PPID = 456
Parent process is terminating.
akhil@DESKTOP-HL1VUUS:~$ ps
  PID TTY          TIME CMD
  456 pts/0        00:00:00 bash
  530 pts/0        00:00:00 ps
```

```

akhil@DESKTOP-HL1VUUS:~$ ps -ef
UID      PID     PPID    C  STIME TTY          TIME CMD
root         1         0  0  04:58 ?        00:00:00 /sbin/init
root         2         1  0  04:58 hvc0     00:00:00 /init
root         6         2  0  04:58 hvc0     00:00:00 plan9 --control-socket 7 --log-level 4
root        49         1  0  04:58 ?        00:00:00 /usr/lib/systemd/systemd-journald
root        97         1  0  04:58 ?        00:00:00 /usr/lib/systemd/systemd-udevd
systemd+   151         1  0  04:58 ?        00:00:00 /usr/lib/systemd/systemd-resolved
systemd+   152         1  0  04:58 ?        00:00:00 /usr/lib/systemd/systemd-timesyncd
root       158         1  0  04:58 ?        00:00:00 /usr/sbin/cron -f -P
message+   159         1  0  04:58 ?        00:00:00 @dbus-daemon --system --address=systemd
root       166         1  0  04:58 ?        00:00:00 /usr/lib/systemd/systemd-logind
root       169         1  0  04:58 ?        00:00:00 /usr/libexec/wsl-pro-service -vv
syslog     176         1  0  04:58 ?        00:00:00 /usr/sbin/rsyslogd -n -iNONE
root       179         1  0  04:58 tty1     00:00:00 /sbin/agetty -o -p -- \u --noclear - li
root       195         1  0  04:58 ?        00:00:00 /usr/bin/python3 /usr/share/unattended-
root       283         2  0  04:59 pts/1    00:00:00 /bin/login -f
akhil      326         1  0  04:59 ?        00:00:00 /usr/lib/systemd/systemd --user
akhil      327        326  0  04:59 ?        00:00:00 (sd-pam)
akhil      349        283  0  04:59 pts/1    00:00:00 -bash
root       454         2  0  05:06 ?        00:00:00 /init
root       455        454  0  05:06 ?        00:00:00 /init
akhil      456        455  0  05:06 pts/0    00:00:00 -bash
root       533         97  0  05:13 ?        00:00:00 (udev-worker)
root       534         97  0  05:13 ?        00:00:00 (udev-worker)
akhil      535        456  0  05:13 pts/0    00:00:00 ps -ef

akhil@DESKTOP-HL1VUUS:~$ ps -ef
UID      PID     PPID    C  STIME TTY          TIME CMD
root         1         0  0  04:58 ?        00:00:00 /sbin/init
root         2         1  0  04:58 hvc0     00:00:00 /init
root         6         2  0  04:58 hvc0     00:00:00 plan9 --control-socket 7 --log-level 4
root        49         1  0  04:58 ?        00:00:00 /usr/lib/systemd/systemd-journald
root        97         1  0  04:58 ?        00:00:00 /usr/lib/systemd/systemd-udevd
systemd+   151         1  0  04:58 ?        00:00:00 /usr/lib/systemd/systemd-resolved
systemd+   152         1  0  04:58 ?        00:00:00 /usr/lib/systemd/systemd-timesyncd
root       158         1  0  04:58 ?        00:00:00 /usr/sbin/cron -f -P
message+   159         1  0  04:58 ?        00:00:00 @dbus-daemon --system --address=systemd
root       166         1  0  04:58 ?        00:00:00 /usr/lib/systemd/systemd-logind
root       169         1  0  04:58 ?        00:00:00 /usr/libexec/wsl-pro-service -vv
syslog     176         1  0  04:58 ?        00:00:00 /usr/sbin/rsyslogd -n -iNONE
root       179         1  0  04:58 tty1     00:00:00 /sbin/agetty -o -p -- \u --noclear - li
root       195         1  0  04:58 ?        00:00:00 /usr/bin/python3 /usr/share/unattended-
root       283         2  0  04:59 pts/1    00:00:00 /bin/login -f
akhil      326         1  0  04:59 ?        00:00:00 /usr/lib/systemd/systemd --user
akhil      327        326  0  04:59 ?        00:00:00 (sd-pam)
akhil      349        283  0  04:59 pts/1    00:00:00 -bash
root       454         2  0  05:06 ?        00:00:00 /init
root       455        454  0  05:06 ?        00:00:00 /init
akhil      456        455  0  05:06 pts/0    00:00:00 -bash
akhil      536        456  0  05:13 pts/0    00:00:00 ps -ef

```

```

akhil@DESKTOP-HL1VUUS:~$ top
top - 05:17:47 up 18 min, 1 user, load average: 0.07, 0.03, 0.00
Tasks: 22 total, 1 running, 21 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7783.6 total, 7166.6 free, 566.1 used, 201.4 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 7217.6 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0   21576   13108  9792 S   0.0   0.2   0:00.34 systemd
    2 root        20   0    3180    2208  2072 S   0.0   0.0   0:00.00 init-systemd(Ub
    6 root        20   0    3196    2132  2008 S   0.0   0.0   0:00.00 init
   49 root        19  -1   50444   15948  14800 S   0.0   0.2   0:00.15 systemd-journal
   97 root        20   0   25020    6372   5172 S   0.0   0.1   0:00.28 systemd-udevd
  151 systemd+   20   0   21344   13220  11096 S   0.0   0.2   0:00.05 systemd-resolve
  152 systemd+   20   0   91036    7952   6984 S   0.0   0.1   0:00.06 systemd-timesyn
  158 root        20   0    4244    2684   2428 S   0.0   0.0   0:00.00 cron
  159 message+   20   0    9640    5332   4636 S   0.0   0.1   0:00.04 dbus-daemon
  166 root        20   0   17984    8856   7812 S   0.0   0.1   0:00.05 systemd-logind
  169 root        20   0 1756104   12268   9848 S   0.0   0.2   0:00.09 wsl-pro-service
  176 syslog     20   0  222516    5976   4600 S   0.0   0.1   0:00.05 rsyslogd
  179 root        20   0    3124    1976   1836 S   0.0   0.0   0:00.00 atetty
  195 root        20   0  107032  23332  13876 S   0.0   0.3   0:00.06 unattended-upgr
  283 root        20   0    6704    4664   3880 S   0.0   0.1   0:00.00 login
  326 akhil       20   0   20320   11572   9452 S   0.0   0.1   0:00.11 systemd
  327 akhil       20   0   21168    3576   1848 S   0.0   0.0   0:00.00 (sd-pam)
  349 akhil       20   0    6080    5340   3676 S   0.0   0.1   0:00.02 bash
  454 root        20   0    3184    1120    980 S   0.0   0.0   0:00.00 SessionLeader
  455 root        20   0    3200    1260   1108 S   0.0   0.0   0:00.04 Relay(456)
  456 akhil       20   0    6076    5396   3676 S   0.0   0.1   0:00.04 bash
  541 akhil       20   0    9464    5808   3512 R   0.0   0.1   0:00.07 top

```

The fork() system call is used to create a new process. The original process is called the parent process, and the newly created process is called the child process. Each process has its own PID, and the child process can get its parent's ID using PPID. During execution, processes move through different states such as running, waiting, and terminated.